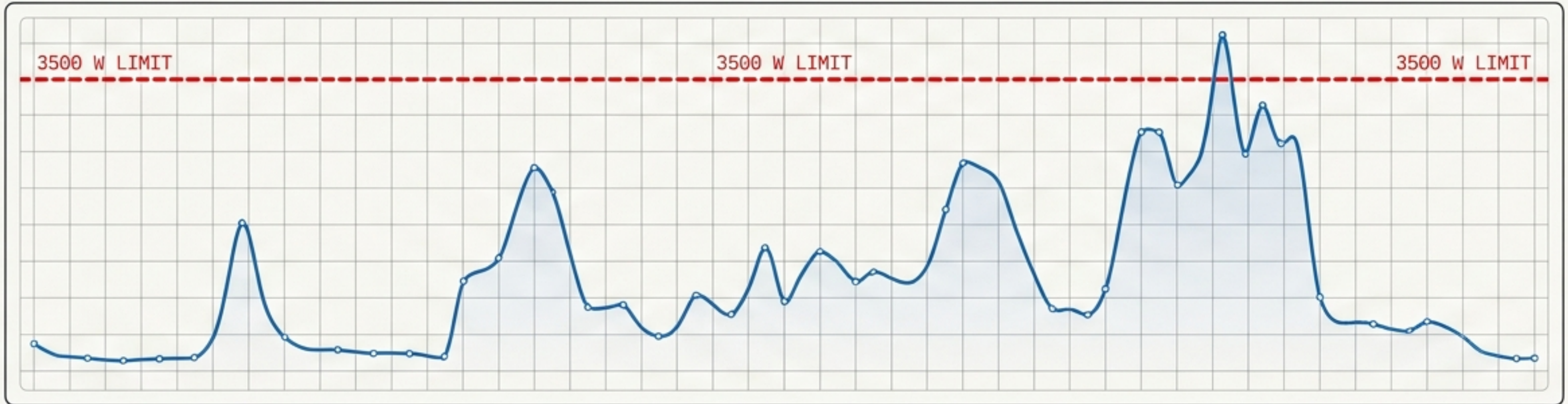
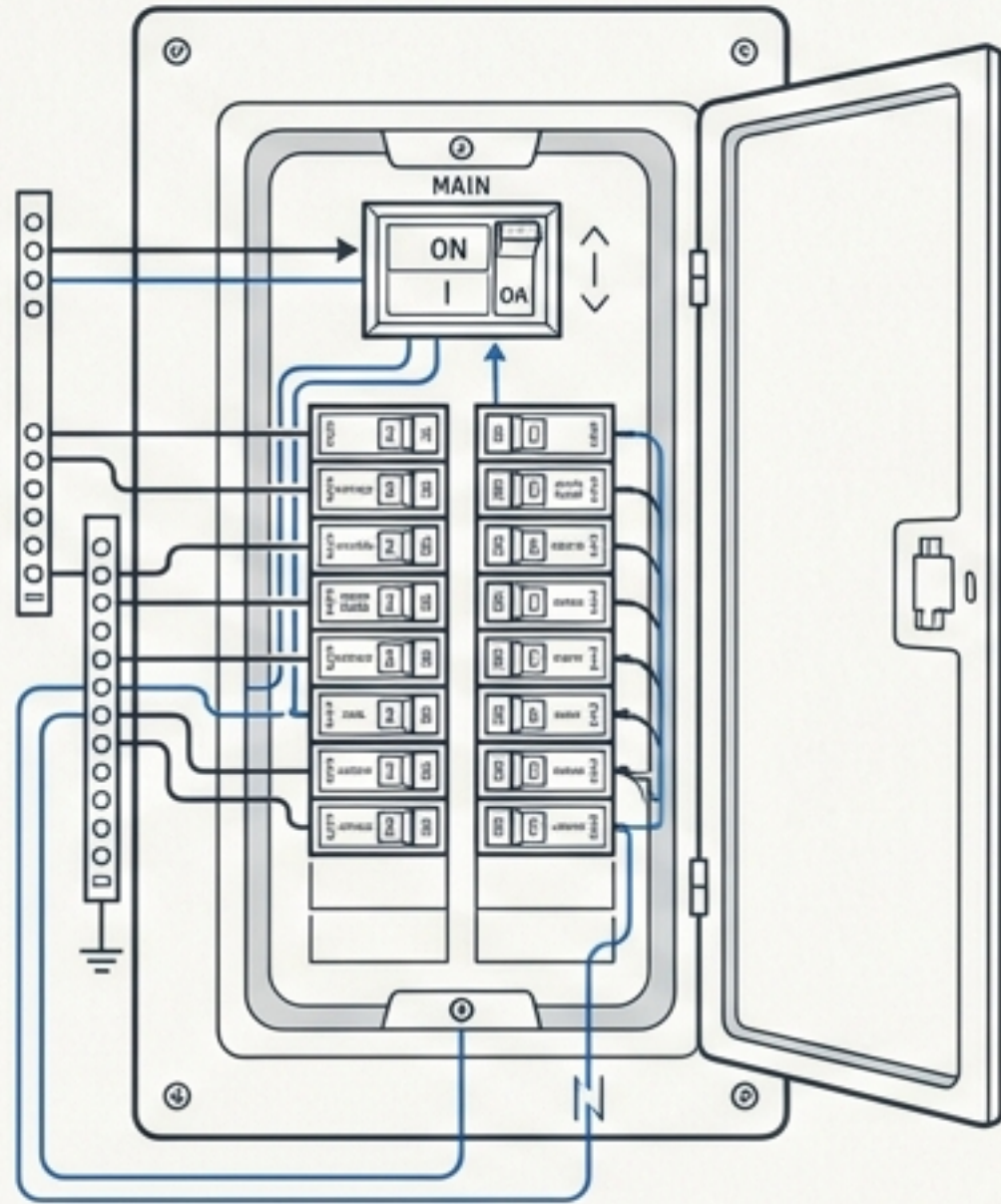


Simulating Household Energy Consumption

A statistical & physics-based code model for load behavior and circuit overload.





SYSTEM CONSTRAINTS

Network:

Main Breaker Rating:

230 V

3500 W

> Goal:

Model household electricity consumption with realistic daily variability.

> The Core Question:

What is the mathematical risk of tripping the main 230V / **3500W** circuit breaker?

$$P_h = P_{\text{lighting}}(h) + P_{\text{fridge}} + P_{\text{washing}}(h) + P_{\text{computer}}(h)$$

Base Load:
Hourly power sum
of all appliances

$$E_{\text{day}} = \frac{\sum_{24}^{24} P_{h,\text{day}}}{1000}$$

Energy Conversion:
Daily consumption in kWh

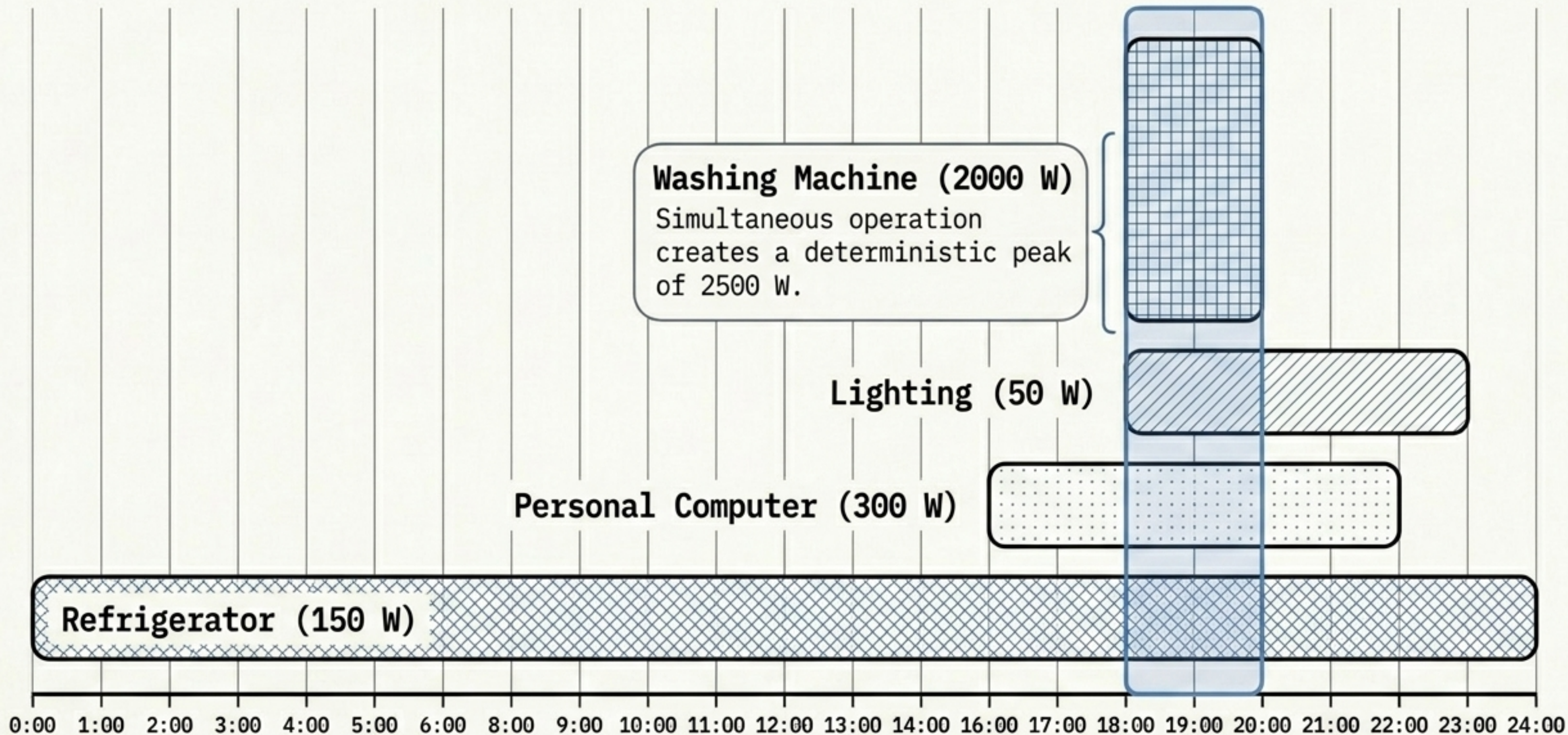
$$P_{\text{peak}} = \max(P_{h,\text{day}})$$

Peak Extraction: Maximum
draw in a 24-hour period

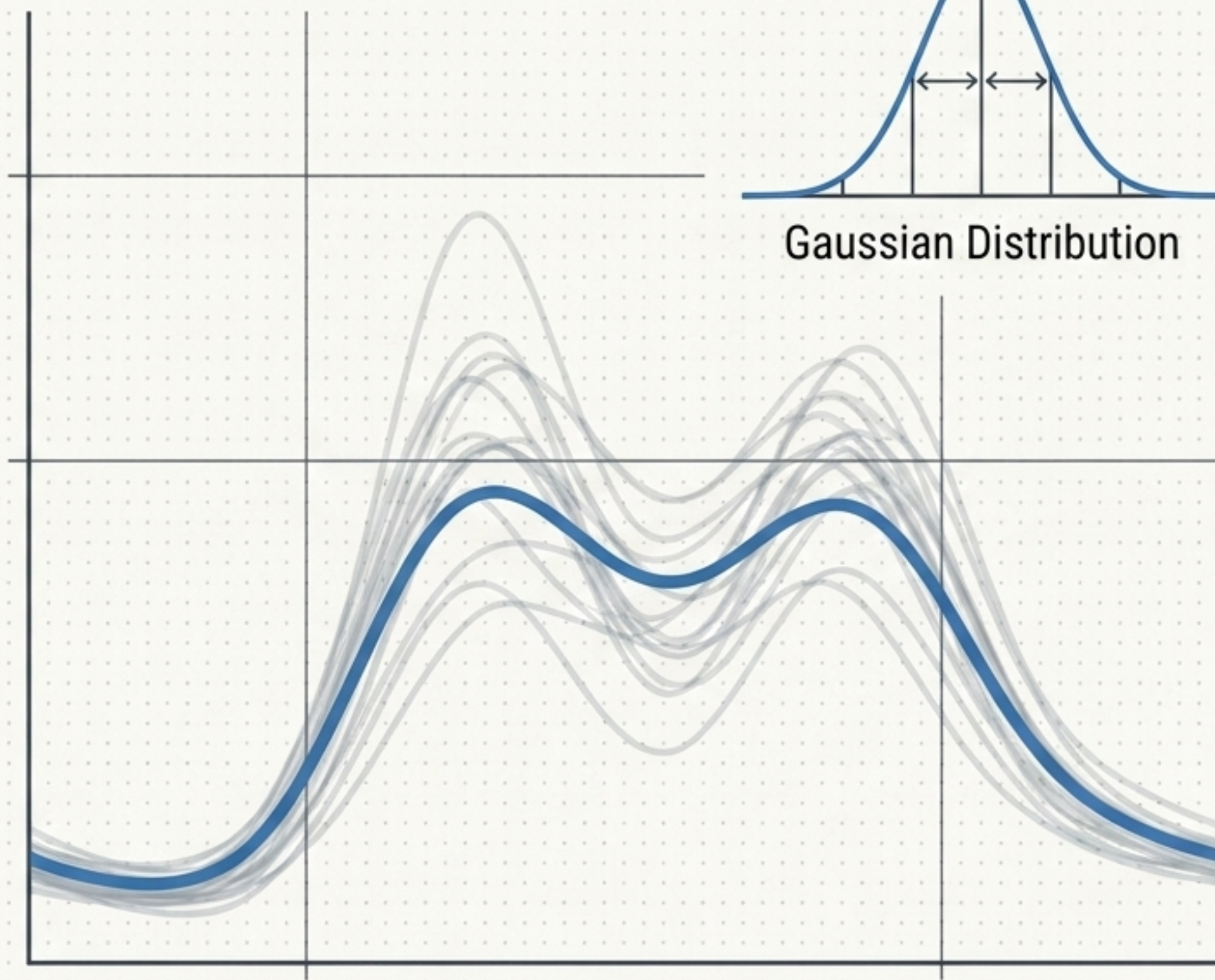
THE OVERLOAD CONDITION:

$$P_{\text{peak}} > 3500 \text{ W}$$

24-HOUR HOUSEHOLD POWER DEMAND



MONTE CARLO METHODOLOGY



MATHEMATICAL LOGIC

$$P_{h,day} = P_h \times v$$

where $v \sim \mathcal{N}(1.0, 0.15)$

Human behavior isn't deterministic. The code applies a $\pm 15\%$ Gaussian variation multiplier over 60 simulated days to model the unpredictability of real-life energy usage.



DETERMINISTIC DAY

Energy

8.18 kWh



Peak Power

2500 W



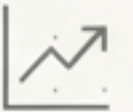
(at 19:00)

60-DAY MONTE CARLO OUTPUT

Mean Energy

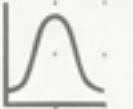
7.99 kWh

(Range: 5.78 - 10.45 kWh)



Mean Peak

2442 W

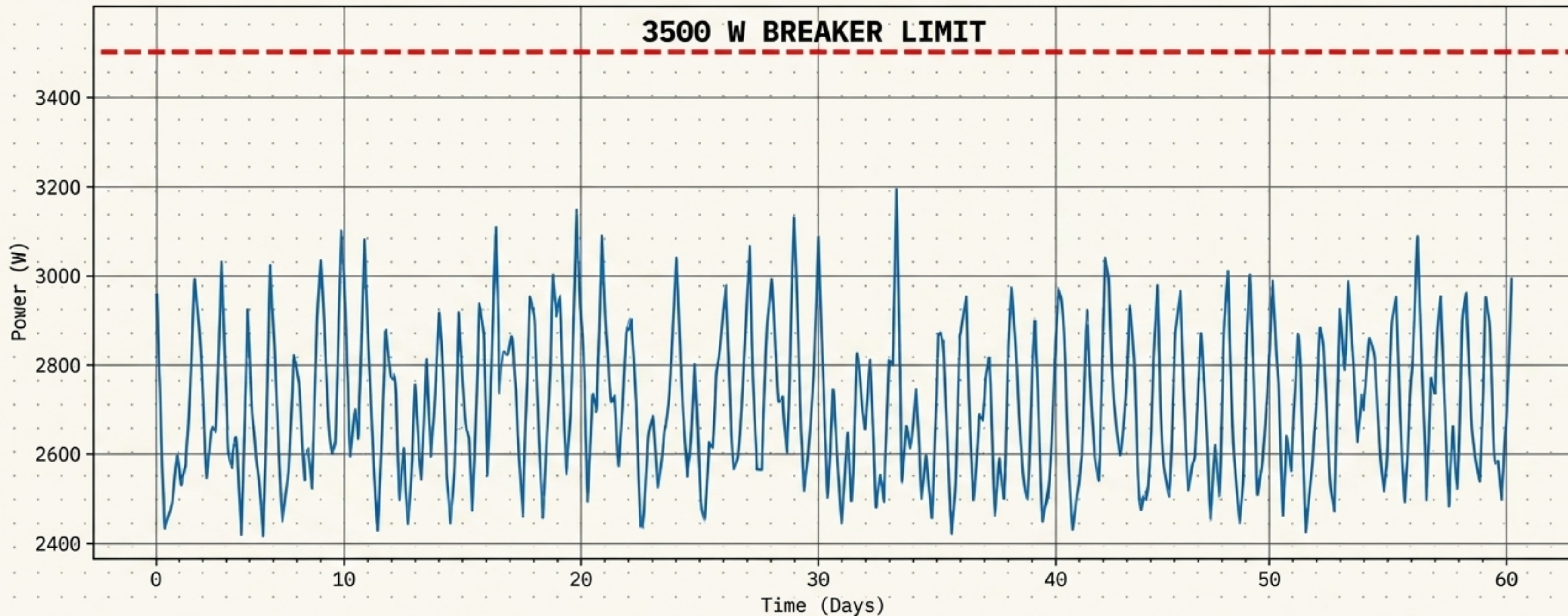


Max Observed Peak

3194.6 W



OVERLOAD PROBABILITY (>3500 W): 0.0%
0 overload days / 60



The Verdict }

The household possesses a significant safety margin. $\pm 15\%$ daily variation heavily impacts total energy use (affecting bills), but barely influences overload risk because the base peak (2500 W) is too low.

{ Alternative Overload Scenarios }

Triggering an overload requires: increasing variation ($\sigma=0.25-0.35$), adding short-duration high-power loads (kettle, oven), or downgrading the breaker to 3000-3200 W.



Resolution Constraints

Hourly resolution obscures dangerous, micro-second power spikes.



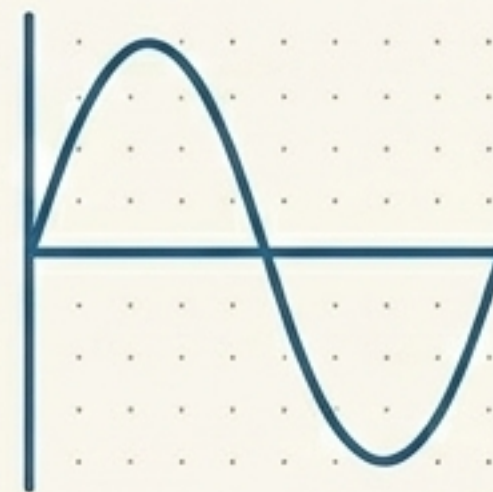
Uniformity

A uniform variation coefficient applied to all loads lacks specific human behavioral modeling.



No Seasonal Dynamics

The model currently ignores weather-dependent loads like air conditioning or electric heating.



Idealized Physics

Omits calculations for reactive power and power factor.



Load Shifting

Move the washing machine cycle to night hours (02:00-05:00) to entirely eliminate the 19:00 convergence peak.



Lighting Upgrades

Replace existing incandescent or halogen bulbs with LED lighting to lower the base evening load.



Future-Proofing

Implement smart plugs, timers, and demand-side management before introducing new high-power appliances like EV chargers or heat pumps.